Slide Set #1

Introduction

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Econometrics
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Notes

What is econometrics?

"*The application of statistical and mathematical methods to the analysis of economic data, with a purpose of giving empirical content to economic theories and verifying them or refuting them*". (Maddala, 1992)

Emphasis on

- Economic theories
- Economic data
- Statistical and mathematical methods

Notes

Typical workflow

- Economic theory (or theories). Example: Keynesian consumption model:
$$C = a + bY$$
- Collect economic data
- Specification of the econometric model:
$$C_i = a + bY_i + \text{random fluctuation}$$
- Inference and/or prediction:
  - testing: $\mathsf{H}_0 : b = 0$ vs $\mathsf{H}_1 : b \neq 0$
  - estimation of the unknowns to do economic analysis: how large is b_{Italy} compared to b_{US}?
  - prediction: if some policy produces $\Delta Y = 10^9$euro, what's the predicted change in C?

Notes

Dependence and causality

Dependence

In terms of probability notions the question is whether $\Pr\{Y \mid X\} \neq \Pr\{Y\}$ or $\Pr\{X \mid Y\} \neq \Pr\{X\}$. Just this. In terms of empirical data we want to know whether variations of Y are somewhat *associated* with variations of X

Causality

Assume that $\Pr\{Y \mid X\} \neq \Pr\{Y\}$, or $\Pr\{X \mid Y\} \neq \Pr\{X\}$, or suppose that data tell us that variations of Y are somewhat associated with variations of X. The question is: why and how this happens? Are we sure that there is a *direct link* between X and Y. Can we exclude that there is an additional Z that is related to both X and Y so that this causes the observed dependence?

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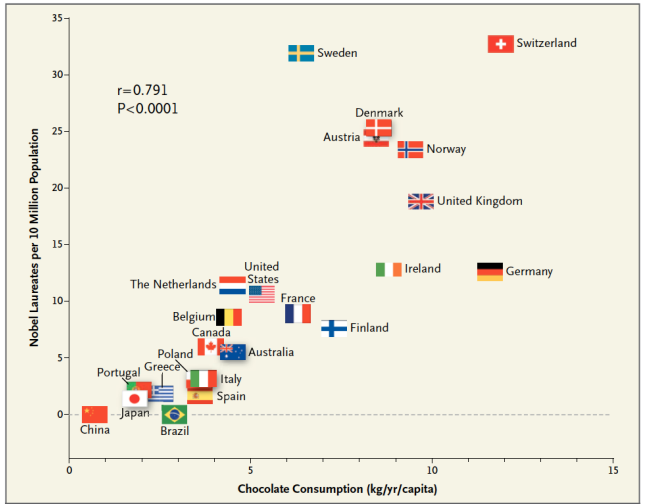


Figure 1. Correlation between Countries' Annual Per Capita Chocolate Consumption and the Number of Nobel Laureates per 10 Million Population.

Source: Messerli, Franz H. 2012. "*Chocolate Consumption, Cognitive Function, and Nobel Laureates*". New England Journal of Medicine. doi:10.1056/NEJMon1211064.

Notes

Predictive modeling

The statistical model "*exploits the dependence*" between an input/feature X and output/outcome variable Y, so that you can predict the outcome value $\hat{Y}_i$ for the ith sample unit for which you only observe the input value X_i.

What's a good model here? Why such prediction works doesn't matter, your problem is to guarantee that the

$$\text{(Squared) Prediction Error} = (\hat{Y}_i - Y_i^{\text{truth}})^2$$

is as small as possible for most sample units.

Predictive modeling is the main paradigm of computer science, artificial intelligence, machine learning, etc.

Notes

Explanatory modeling

The statistical model "*explains and describes*" the link based on which an exogenous/independent variable X determines variations of an endogenous/dependent variable Y.

Usually a theoretical economic model itself provides the causality link.

What's a good model? We look for different requirements here:

- causal parameter(s) of the model, i.e. the parameters linking the X to the Y, uniquely describe the impact of X on Y (indentifiability)
- the model is valid in the sense that the underlying causal hypothesis cannot be rejected based on the empirical data (testing)
- there is a way to use empirical data to estimate the causal parameter(s) of the model uniquely and accurately (bias, consistency, etc.)

Notes

Example: the Keynes' consumption model would be good if

- b "*uniquely*" explains the impact of Y on C
- we can reject $\mathsf{H}_0 : b = 0$ against $\mathsf{H}_1 : b \neq 0$
- there is a way to produce a unique estimate $\hat{b}$ such that

$$\text{Estimation Error} = \hat{b} - b^{\text{truth}}$$

  is small enough in some sense

Explanatory modeling is the main paradigm in econometrics and other social sciences.

In econometrics *Explanatory models* are subdivided in *structural models* and *reduced form models*

Notes

Mixing the paradigms

The two paradigms have different goals, although they share a lot of statistical techniques

⚠ Statistical methods need to be used and interpreted in a different way depending on whether we want to *predict* or *explain*

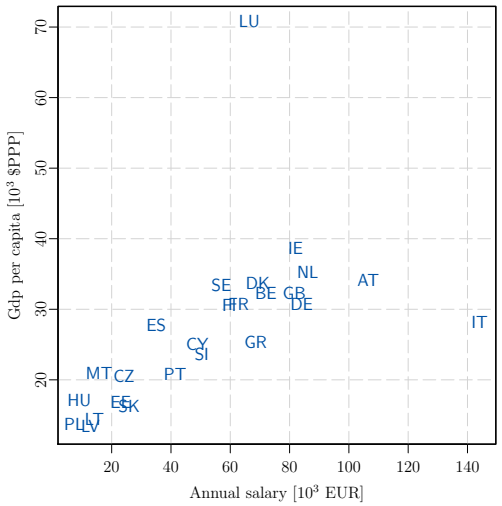
There are fields where traditionally the distinction between these two paradigms is smooth (epidemiology, bio-sciences, etc).

⚠ The modern abundance of massive data collections increased the popularity of the predictive modeling approach. There is a recent tendency to mix the two paradigms unconsciously, but this can easily lead to dramatically wrong scientific statements

Notes

Linear modeling

European parliament salaries



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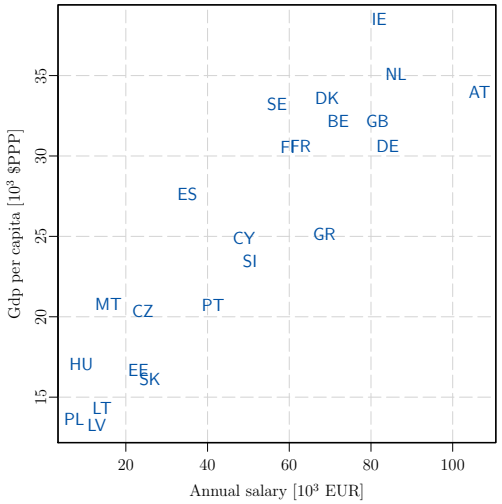
After cleaning outlying observations



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Joint distribution, covariance and correlation

In previous courses you learned that

- a pair of random variables (X, Y) are linearly dependent (statistical notion) if their joint distribution is such that

$$\begin{aligned}\mathrm{Cov}(X,Y) &= \mathrm{E}[(X-\mu_X)(Y-\mu_X)] \neq 0 \\ &= \iint (x-\mu_X)(y-\mu_Y)\, f_{X,Y}(x,y)\, dx\, dy \neq 0\end{aligned}$$

- the strength of the dependence is measured by the correlation

$$\mathrm{Cor}[X,Y] = \frac{\mathrm{Cov}[X,Y]}{\sqrt{\mathrm{Var}[X]}\sqrt{\mathrm{Var}[Y]}} \in [-1,1]$$

- independence $\implies \mathrm{Cor}[X,Y] = 0$, the converse is not true
- $\mathrm{Cor}[X,Y] = \mathrm{Cor}[Y,X]$ the symmetry of the covariance operator doesn't allow to make causal statements

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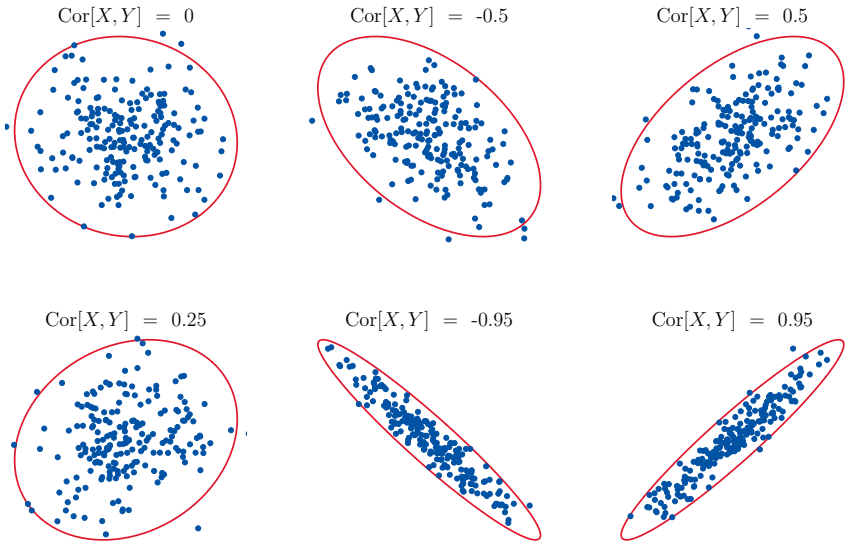

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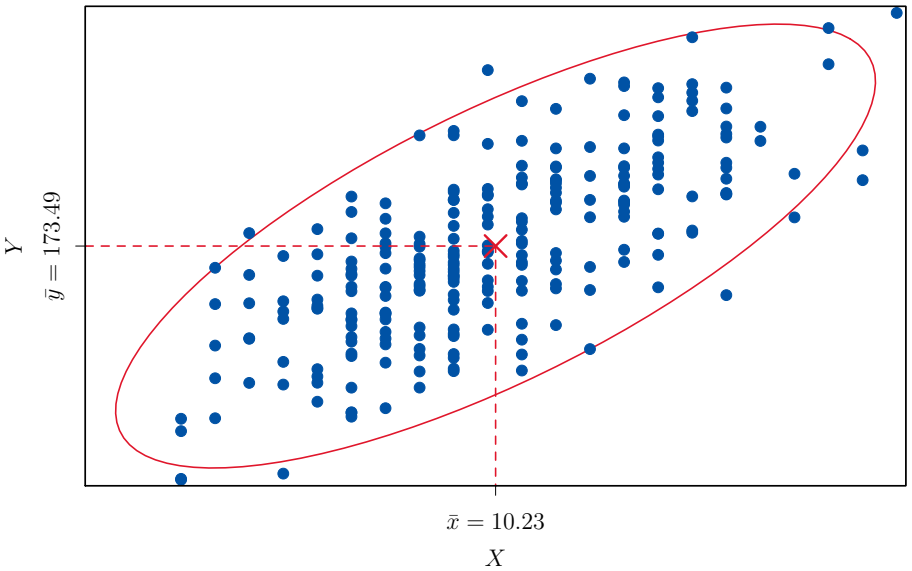

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If (X, Y) are correlated, a sample from their joint distribution most of the times produces a scatter where the majority of the points lie in an ellipsoidal region centered at sample means $(\bar{x}, \bar{y})$

The volume of the ellipse captures the overall joint dispersion (multivariate variance)

Highly correlated pairs ($\mathrm{Cor}[X, Y]$ close to ± 1) have scatters compressed along the main axes of the ellipses

How do we model this? A statistical model for (X, Y) needs to be able to reproduce this behavior of the joint distribution. What kind of sampling design can reproduce such a thing?

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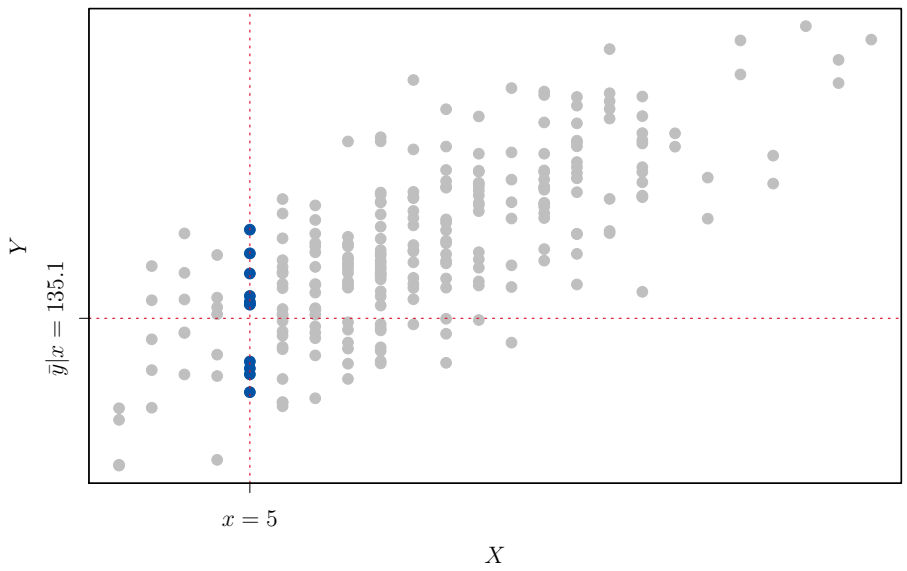

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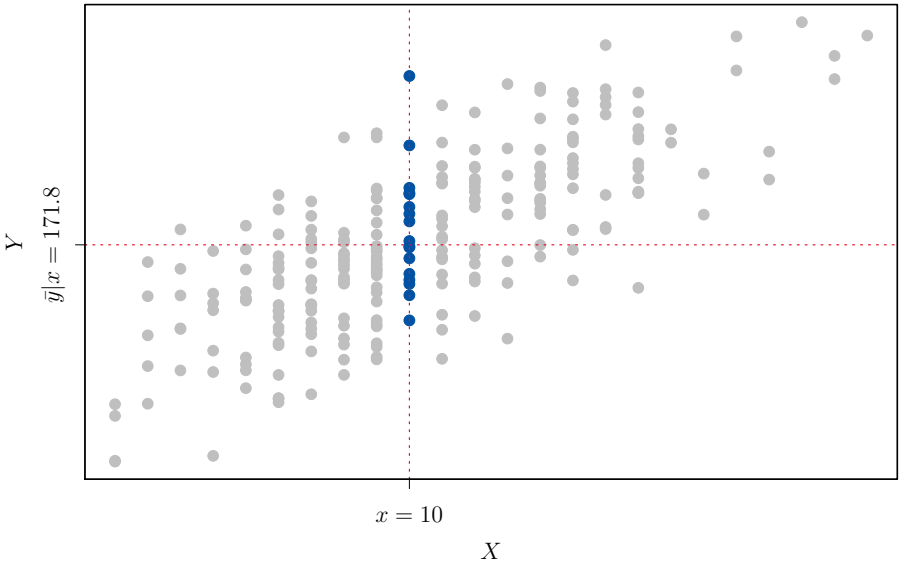

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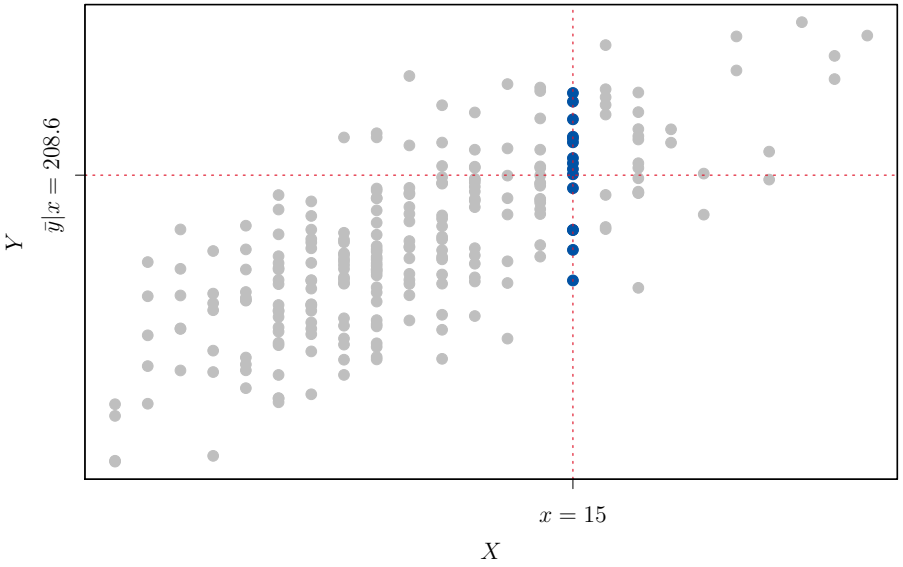

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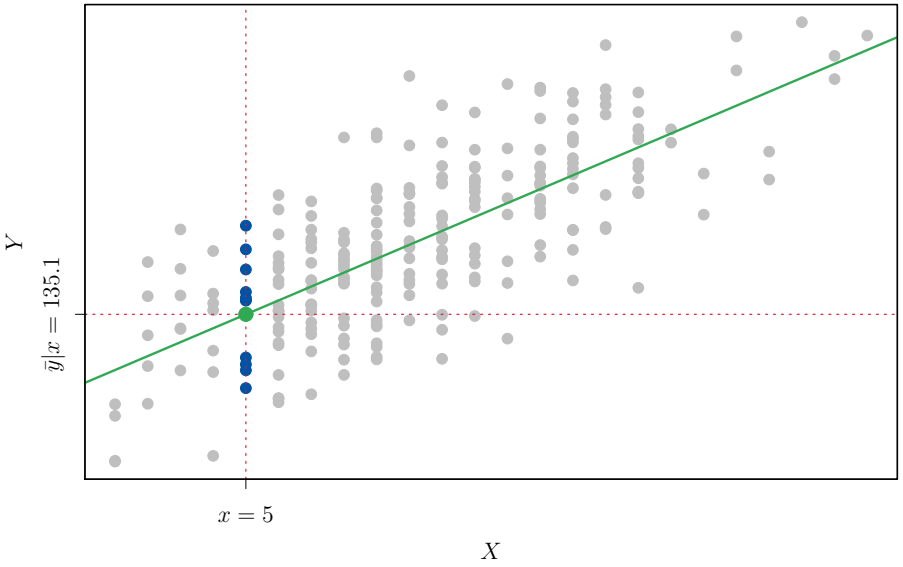

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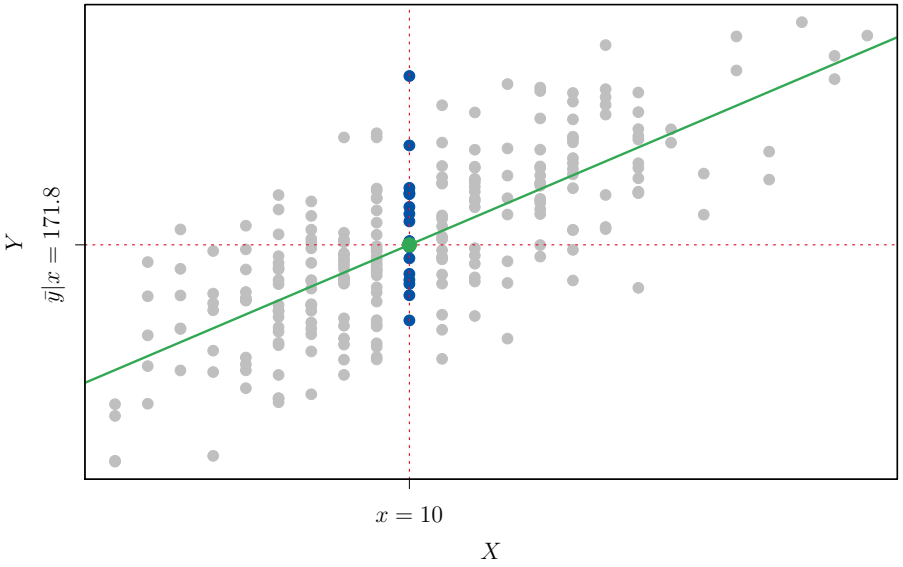

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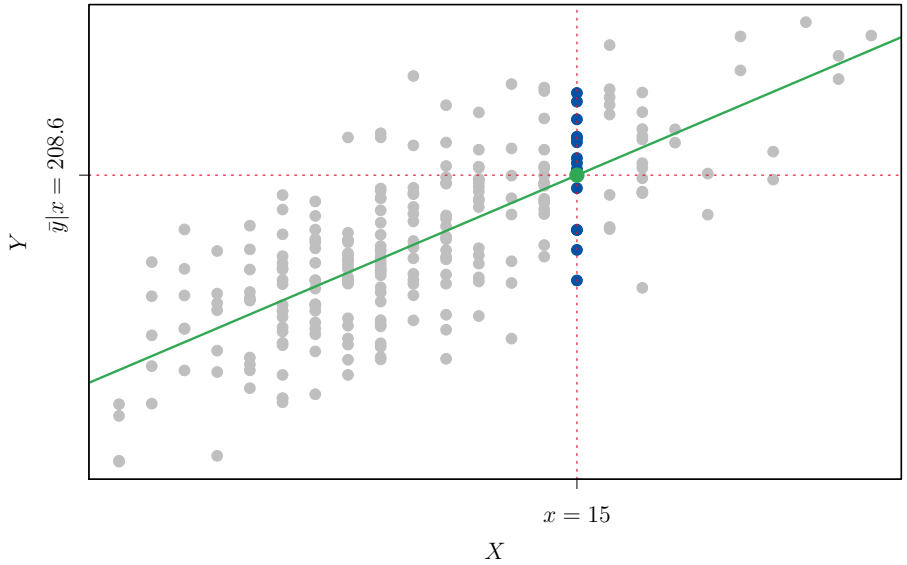

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The conditional mean of $Y \mid X$ *increases proportionally* as we increase X

For a fixed $X = x$, the points are randomly scattered around the conditional mean of $Y \mid X = x$

Therefore if (Y_i, X_i) are pairs of random variables sampled from a joint distribution for which $\mathrm{Cor}[X, Y] \neq 0$, a model to represent what we would observe from such distribution would be

$$\begin{aligned} Y_i &= \mathrm{E}[Y|X_i] + \text{random fluctuation} \\ &= \beta_0 + \beta_1 X_i + \text{random fluctuation} \end{aligned}$$

The latter is a linear regression model. It will be the object of interest of this course. It allows to reproduce the observed correlation. This can be used to *predict* or to *explain*

Notes

Correlation twists in modern big-data

Big data = massive collection of data with huge dimensions in both

- n =number of sample units
- p =number of variables/features measured on each unit

Issues relevant for econometric applications

- n much smaller than p
- spurious dependence
- heterogeneity

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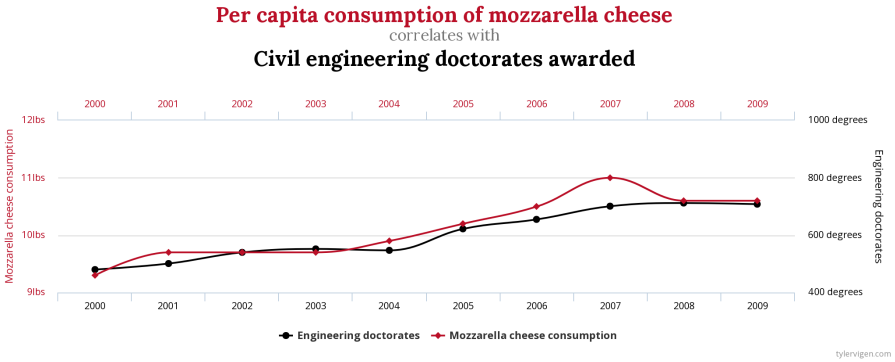

source: `http://tylervigen.com/spurious-correlation`

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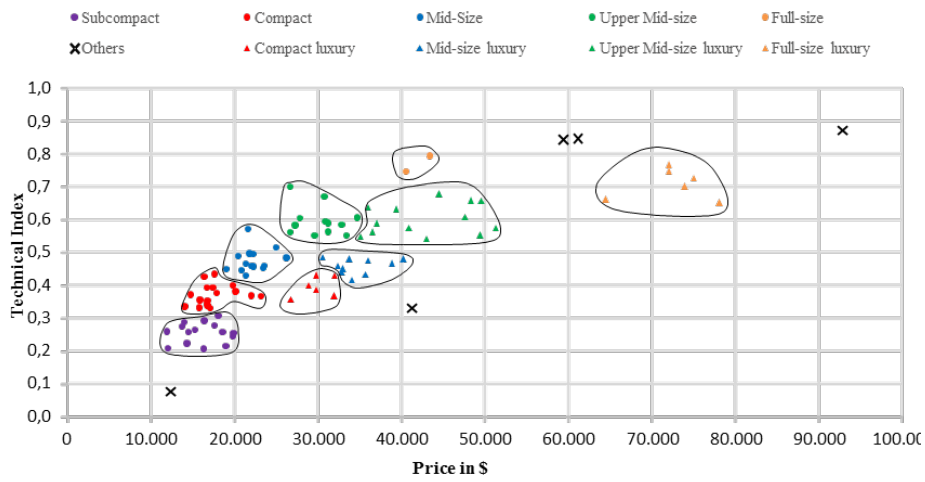

source: `https://www.allaboutlean.com/automotive-market-strategy/`

Notes

Collecting data

We are flooded with data

"Good" econometric analysis crucially depends on the availability of "good" economic data. Key ingredients are: a) good data sources; b) appropriate data processing.

In modern days the ability to perform sophisticated data (pre)processing tasks is essential. Computing is a necessary tool for modern econometrics.

Major issues

- Relevant economic variables are not always observable (e.g. expectations)
- Wide gap between the ideal measure and its observable counterpart (e.g. stock of capital in production functions)
- Data acquisition frequency is often not appropriate
- Non-response can be serious if respondents are not a random sample drawn from the population ("selectivity bias" problem)

Notes

Types of data vs indexing of the units

Suppose $X =$`GDP`

- Cross-section: X is measured on the ith unit at a given time point, i is a country, an individual, an household, etc.
  Example: $i = 1 =$`UK`, $i = 2 =$`Italy`, X_1 GDP of UK in 2018, $X_2 =$GDP of Italy in 2018
- Time series: X is measured on a certain unit at a time point i, i can be a year, month, quarter, etc.
  Example: $X_{1960} =$ GDP of Italy in 1960, $X_{1961} =$ GDP of Italy in 1961, etc.
- Panel or longitudinal data: i, j are time and unit indexes. X is measured on the ith unit in the jth period.
  Example: $X_{1,1980}$ is the GDP of UK in 1980, $X_{2,2000}$ is the GDP of Italy measured in 2000.

Notes

Types of variables vs levels/labels

Level/label: value recorded for each sample unit.
Example: $X_i = 100$, 100 is the level, or $G_i =$`Male`, `Male` is the label/level

- Quantitative (aka numerical) variables: levels are numerical, differences between levels have a metric interpretation. Numerical variables can be: continuous (e.g. income), discrete (e.g. number of patents), and discretized continuous variables (e.g. age)
- Ordinal variables: levels can be ordered, differences between levels don't have a metric interpretation.
  Example: $Y =$`income`$=\{$ `upper`$=3$, `middle`$=2$, `lower`$=1\}$, therefore `lower`$\preceq$`upper`, but *upper-lower* = 3-1 = 2 = ?
- Categorical (aka nominal) variables: labels cannot be ordered, differences between labels don't have a metric interpretation.
  Example: $S=$`sex`$=\{0=$`Male`, $1=$`Female`$\}$, can we say `female`$\preceq$`male` or `male`$\preceq$`female`? Moreover, *Female -Male* = 1-0 = 1 = ?

Notes